

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter Examination – 2022

Course: B. Tech.

Branch :

Semester :III

Subject Code & Name: Engineering Mathematics(BTBS301)

Max Marks: 60

Date: / /20

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) Find the Laplace transform of $e^{-2t} \int_0^t \frac{\cos 2t}{t} dt$.	Understand	6
B) Find the Laplace transform of the periodic function, $f(t) = \frac{t}{T}$ for $0 < t < T$, & $f(t + T) = f(t)$.	Understand	6
C) By using Laplace transform, evaluate $\int_0^\infty e^{-2t} t^2 \sin 3t dt$	Evaluation	6
Q.2 Solve Any Two of the following.		12
A) By using convolution theorem, find inverse Laplace transform of $\frac{s}{(s^2+1)(s^2+4)}$	Application	6
B) Find inverse Laplace transform of $\cot^{-1}(\frac{s+3}{2})$	Application	6
C) Using Laplace Transform, solve $\frac{d^2y}{dt^2} - 4 \frac{dy}{dt} + 3y = e^{-t}$, given $y(0) = 1$ & $y'(0) = 0$	Application	6
Q.3 Solve Any Two of the following.		12
A) Using the Fourier integral representation, show that i) $\int_0^\infty \frac{\omega \sin x \omega}{1+\omega^2} d\omega = \frac{\pi}{2} e^{-x} (x > 0)$ ii) $\int_0^\infty \frac{\cos x \omega}{1+\omega^2} d\omega = \frac{\pi}{2} e^{-x} (x \geq 0)$	Understand	6

Find the Fourier transform of the function			
b)	$f(x) = \begin{cases} 1 - x^2 & x \leq 1 \\ 0 & x > 1 \end{cases}$ Hence evaluate $\int_0^{\infty} \frac{x \cos x - \sin x}{x^3} \cos \frac{x}{2} dx$	Evaluation	6
c)	Using Parseval's identity, show that $\int_0^{\infty} \frac{t^2}{(4+t^2)(9+t^2)} dt = \frac{\pi}{10}$	Application	6
Q.4 Solve Any Two of the following.			12
A)	Form the partial differential equation by eliminating arbitrary function from $f(x + y + z, x^2 + y^2 + z^2) = 0$	Understand	6
B)	Solve the partial differential equation $(x^2 - yz)p + (y^2 - zx)q = z^2 - xy$	Application	6
C)	Find the temperature in a bar of length 2 units whose ends are kept at zero temperature & lateral surface insulated if the initial temperature is $\sin \frac{\pi x}{2} + 3 \sin \frac{5\pi x}{2}$	Application	6
Q.5 Solve Any Two of the following. https://www.batuonline.com			12
A)	Prove that the function $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$ is a harmonic function & hence determine the corresponding analytic function, $f(z) = u + iv$.	Understand	6
B)	Evaluate, by using Cauchy's integral formula: i) $\oint_C \frac{e^{-z}}{z+1} dz$, where C is the circle $z = 2$. ii) $\oint_C \frac{\sin^2 z}{(z-\frac{\pi}{6})^3} dz$, where C is the circle $z = 1$.	Evaluation	6
C)	Evaluate $\int_C \frac{2z-1}{z(z+1)(z-3)} dz$, where C is the circle $z = 2$	Evaluation	6
*** End ***			

Model Answer

Q.3
A]

$$\mathcal{L}\{\cos 2t\} = \frac{s}{s^2 + 2^2}$$

$$\mathcal{L}\left\{\frac{\cos 2t}{t}\right\} = \int_s^\infty \frac{s}{s^2 + 2^2} ds$$

$$= \frac{1}{2} \int_s^\infty \frac{2s}{s^2 + 2^2} ds$$

$$= \frac{1}{2} \left[\log(s^2 + 4) \right]_s^\infty$$

$$= \frac{1}{2} [\log \infty - \log(s^2 + 4)]$$

$$\mathcal{L}\left\{\frac{\cos 2t}{t}\right\} = -\frac{1}{2} \log(s^2 + 4)$$

$$\mathcal{L}\left\{\int_0^t \frac{\cos 2t}{t} dt\right\} = \frac{1}{s} \left[-\frac{1}{2} \log(s^2 + 4) \right]$$

$$\mathcal{L}\left\{e^{-2t} \int_0^t \frac{\cos 2t}{t} dt\right\} = \frac{1}{s+2} \left[-\frac{1}{2} \log((s+2)^2 + 4) \right]$$

$$= -\frac{1}{2(s+2)} \log(s^2 + 2s + 4 + 4)$$

$$= -\frac{1}{2(s+2)} \log(s^2 + 2s + 8)$$

B) $f(t) = \frac{t}{T} \quad 0 \leq t \leq T$

Here $L\{f(t)\} = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t) dt$

$$= \frac{1}{1-e^{-sT}} \int_0^T e^{-st} \left(\frac{t}{T}\right) dt$$

$$= \frac{1}{T(1-e^{-sT})} \int_0^T e^{-st} t dt$$

$$= \frac{1}{T} \frac{1}{1-e^{-sT}} \left[-\frac{T}{s} e^{-st} - \frac{e^{-st}}{s^2} + \frac{1}{s^2} \right]$$

$$= \frac{1}{T} \frac{1}{1-e^{-sT}} \left[-\frac{T}{s} e^{-sT} + \frac{1}{s^2} (1 - e^{-sT}) \right]$$

$$= -\frac{1}{s} \left(\frac{e^{-sT}}{1-e^{-sT}} \right) + \frac{1}{s^2 T}$$

$$L\{f(t)\} = \frac{1}{s^2 T} - \frac{1}{s} \left(\frac{e^{-sT}}{1-e^{-sT}} \right)$$

$$\boxed{7} \quad \int_0^{\infty} e^{-2t} t^2 \sin 3t \, dt.$$

We have

$$\int_0^{\infty} e^{-st} f(t) \, dt = L\{f(t)\}$$

$$\int_0^{\infty} e^{-2t} (t^2 \sin 3t) \, dt = L\{t^2 \sin 3t\}$$

$$s = 2.$$

Now,

$$L\{\sin 3t\} = \frac{3}{s^2 + 3^2}.$$

Next,

$$L\{t^2 \sin 3t\} = (-1)^2 \frac{d^2}{ds^2} \left\{ \frac{3}{s^2 + 9} \right\}, \quad s = 2$$

$$= (1) \frac{d}{ds} \left\{ \frac{d}{ds} \frac{3}{s^2 + 9} \right\}, \quad s = 2$$

$$= \frac{d}{ds} \left\{ -\frac{3}{s^2 + 9} \times 2s \right\}, \quad s = 2$$

$$= -6 \frac{d}{ds} \left\{ \frac{s}{(s^2 + 9)^2} \right\}, \quad s = 2.$$

$$-6 \left[\frac{(s^2+9)^2(1) - s(2(s^2+9) \times 2s)}{(s^2+9)^4} \right], s=2$$

$$= -6 \left[\frac{s^2+9-4s^2}{(s^2+9)^4} \right], s=2$$

$$= -6 \left[\frac{s^2+9-4s^2}{(s^2+9)^3} \right], s=2$$

$$= -6 \left[\frac{-3s^2+9}{(s^2+9)^3} \right], s=2$$

$$= -6 \left[\frac{-3(2)^2+9}{(2^2+9)^3} \right]$$

$$= -6 \left[\frac{-12+9}{(4+9)^3} \right]$$

$$= \frac{-6(-3)}{(13)^3}$$

$$= \frac{18}{(13)^3}$$

$$= \frac{18}{2197}$$

$$\int_0^{\infty} e^{-2t} [t^2 \sin 3t] dt = \frac{18}{2197}$$

Q.2

A. Here $\bar{f}(s) = \frac{s}{(s^2+1)(s^2+4)} = \frac{1}{s^2+1} \cdot \frac{s}{s^2+4} = \bar{f}_1(s) \cdot \bar{f}_2(s)$

When $\bar{f}_1(s) = \frac{1}{s^2+1} \Rightarrow \mathcal{L}^{-1}\{\bar{f}_1(s)\} = f_1(t) = \mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} = \sin t$

$\bar{f}_2(s) = \frac{s}{s^2+2^2} \Rightarrow \mathcal{L}^{-1}\{\bar{f}_2(s)\} = f_2(t) = \mathcal{L}^{-1}\left\{\frac{s}{s^2+4}\right\} = \cos 2t$

Now.

$f_1(t) = \sin t \Rightarrow f_1(u) = \sin u$

$f_2(t) = \cos 2t \Rightarrow f_2(t-u) = \cos 2(t-u) = \cos(2t-2u)$

Hence, by the convolution theorem,

$$\mathcal{L}^{-1}\{\bar{f}(s)\} = \mathcal{L}^{-1}\{\bar{f}_1(s) \bar{f}_2(s)\} = \int_0^t f_1(u) f_2(t-u) du = \int_0^t \sin u \cos(2t-2u) du$$

$$= \frac{1}{2} \int_0^t 2 \sin u \cos(2t-2u) du$$

$$= \frac{1}{2} \int_0^t \{\sin(u+2t-2u) + \sin(3u-2u)\} du$$

$$= \frac{1}{2} \int_0^t \{\sin(2t-u) + \sin(3u-2u)\} du$$

$$= \frac{1}{2} \left[\{-\cos t + \cos 2t\} + \left\{-\frac{\cos t}{3} + \frac{\cos 2t}{3}\right\} \right]$$

$$= \frac{1}{2} \left[\frac{2}{3} \cos t - \frac{2}{3} \cos 2t \right]$$

$$= \frac{1}{3} (\cos t - \cos 2t)$$

$$f(t) = \frac{1}{3} (\cos t - \cos 2t)$$

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$$f(s) = \cot' \left(\frac{s+3}{2} \right)$$

$$\text{gives, } \frac{d\bar{f}(s)}{ds} = - \frac{1}{1 + \left(\frac{s+3}{2} \right)^2} \left(\frac{1}{2} \right)$$

$$= - \left\{ \frac{2}{(s+3)^2 + (2)^2} \right\}$$

$$\Rightarrow \bar{f}' \left\{ \frac{d\bar{f}(s)}{ds} \right\} = - \bar{f}' \left\{ \frac{2}{(s+3)^2 + (2)^2} \right\}$$

$$\Rightarrow -t f(t) = -e^{3t} \bar{f}' \left\{ \frac{2}{s^2 + 2^2} \right\}$$

$$-t f(t) = -e^{3t} \sinh 2t$$

$$f(t) = \frac{e^{3t} \sinh 2t}{-t}$$

$$f(t) = \frac{e^{-3t} \sinh 2t}{t}$$

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given

$$\frac{d^2 y}{dt^2} - 4 \frac{dy}{dt} + 3y = e^t, \text{ \& } y(0) = 1 \& y'(0) = 0.$$

The given diffⁿ eqⁿ can be re-written as.

$$y''(t) - 4y'(t) + 3y(t) = e^t; y(0) = 1, y'(0) = 0$$

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$$\Rightarrow \{y''(t)\} - 4\{y'(t)\} + 3\{y(t)\} = \{e^{-t}\}.$$

$$\Rightarrow s^2 y(s) - sy(0) - y'(0) - 4\{sy(s) - y(0)\} + 3y(s) = \frac{1}{s+1}$$

$$s^2 y(s) - s(1) - 0 - 4\{sy(s) - 1\} + 3y(s) = \frac{1}{s+1}$$

$$\Rightarrow s^2 y(s) - s - 4sy(s) + 4 + 3y(s) = \frac{1}{s+1}$$

$$\Rightarrow (s^2 - 4s + 3)y(s) = s - 4 + \frac{1}{s+1}$$

$$\Rightarrow (s-1)(s-3)y(s) = s - 4 + \frac{1}{s+1}$$

$$\Rightarrow y(s) = \frac{s-4}{(s-1)(s-3)} + \frac{1}{(s-1)(s-3)(s+1)}$$

$$= \frac{(s-4)(s+1) + 1}{(s-1)(s-3)(s+1)}$$

$$= \frac{s^2 - 3s - 3}{(s-1)(s-3)(s+1)}$$

$$\text{Let } \frac{s^2 - 3s - 3}{(s-1)(s-3)(s+1)} = \frac{A}{s-1} + \frac{B}{s-3} + \frac{C}{s+1} \quad \text{--- (2)}$$

$$\Rightarrow s^2 - 3s - 3 = A(s-3)(s+1) + B(s-1)(s+1) + C(s-1)(s-3)$$

Putting $s=1$
 ~~$s=1$~~ we have

$$(1)^2 - 3(1) - 3 = A(-2)(2)$$

$$-5 = -4A$$

$$A = \frac{5}{4}$$

$$\boxed{A = \frac{5}{4}}$$

put $s=3$

$$(3)^2 - 3(3) - 3 = B(2)(4) \Rightarrow B = -\frac{3}{8}$$

Putting $s=-1$,

$$(-1)^2 - 3(-1) - 3 = C(-2)(-4) \Rightarrow C = 1/8$$

From (2)

$$y(s) = \frac{5}{4} \left\{ \frac{1}{s-1} \right\} - \frac{3}{8} \left\{ \frac{1}{s-3} \right\} + \frac{1}{8} \left\{ \frac{1}{s+1} \right\}$$

$$\Rightarrow \mathcal{L}^{-1}\{y(s)\} = \frac{5}{4} \mathcal{L}^{-1}\left\{ \frac{1}{s-1} \right\} - \frac{3}{8} \mathcal{L}^{-1}\left\{ \frac{1}{s-3} \right\} + \frac{1}{8} \mathcal{L}^{-1}\left\{ \frac{1}{s+1} \right\}$$

$$y(t) = \frac{5}{4} e^t - \frac{3}{8} e^{3t} + \frac{1}{8} e^{-t}$$

$$\boxed{y(t) = \frac{5}{4} e^t - \frac{3}{8} e^{-3t} + \frac{1}{8} e^{-t}}$$

Q 3.

A. The Fourier integral representation is given by

$$f(x) = \int_0^{\infty} B(\lambda) \sin \lambda x \, d\lambda \quad \text{--- (1)}$$

$$\text{when } B(\lambda) = \frac{2}{\pi} \int_0^{\infty} f(x) \sin \lambda x \, dx$$

$$= \frac{2}{\pi} \int_0^{\infty} e^{-x} \sin \lambda x \, dx \quad \text{if } f(x) = e^{-x}$$

$$= \frac{2}{\pi} \left[\frac{e^{-x}}{1+\lambda^2} (-\sin \lambda x - \lambda \cos \lambda x) \right]_0^{\infty}$$

$$= \frac{2\lambda}{\pi(1+\lambda^2)}$$

From (1)

$$e^{-x} = \frac{2}{\pi} \int_0^{\infty} \frac{\lambda \sin \lambda x}{1+\lambda^2} \, d\lambda$$

$$\int_0^{\infty} \frac{\lambda \sin \lambda x}{1+\lambda^2} \, d\lambda = \frac{\pi}{2} e^{-x}$$

$$\Rightarrow \int_0^{\infty} \frac{w \sin wx}{1+w^2} \, dw = \frac{\pi}{2} e^{-x} \quad (x > 0)$$

(11) The Fourier cosine representation of $f(x)$ is

$$f(x) = \int_0^{\infty} A(\lambda) \cos \lambda x \, d\lambda$$

$$\text{when } A(\lambda) = \frac{2}{\pi} \int_0^{\infty} f(x) \cos \lambda x \, dx$$

$$= \frac{2}{\pi} \int_0^{\infty} e^{-x} \cos \lambda x \, dx \quad \text{if } f(x) = e^{-x}$$

$$= \frac{2}{\pi} \left[\frac{e^{-x}}{1+j^2} (-1 \cos xn + 1 \sin xn) \right]_0^{\infty}$$

$$= \frac{2}{\pi (1+j^2)}$$

From (2) we obtain

$$e^{-x} = \frac{2}{\pi} \int_0^{\infty} \frac{\cos xn}{1+j^2} dj$$

$$\Rightarrow \int_0^{\infty} \frac{\cos xn}{1+j^2} dj = \frac{\pi}{2} e^{-x}$$

$$\Rightarrow \int_0^{\infty} \frac{\cos xw}{1+w^2} dw = \frac{\pi}{2} e^{-x} \quad (x \geq 0)$$

B) The Fourier transform of $f(x)$ is given by

$$F(s) = \int_{-\infty}^{\infty} f(x) e^{isx} dx$$

$$= \int_{-1}^1 (1-x^2) e^{isx} dx$$

$$= \int_{-1}^1 (1-x^2) (\cos sx + i \sin sx) dx$$

$$= \int_{-1}^1 (1-x^2) \cos sx dx + i \int_{-1}^1 (1-x^2) \sin sx dx$$

$$= 2 \int_0^1 (1-x^2) \cos sx dx \quad [1-x^2 \text{ is even fun}]$$

$$= -\frac{4}{s^3} (s \cos s - \sin s)$$

Now, by inversion formula, we obtain.

$$f(n) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(s) e^{isn} ds.$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} -\frac{4}{s^3} (s \cos s - s i h s) (\cos sn - i s i h sn) ds.$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} -\frac{4}{s^3} (s \cos s - s i h s) \cos sn ds.$$

$$\text{Put } x = \frac{s}{2} \Rightarrow f(n) = 3/4$$

$$-\frac{4}{\pi} \int_0^{\infty} \left(\frac{s \cos s - s i h s}{s^3} \right) \cos\left(\frac{s}{2}\right) ds = \frac{3}{4}$$

$$\Rightarrow \int_0^{\infty} \left(\frac{x \cos x - s i h x}{x^3} \right) \cos\left(\frac{x}{2}\right) dx = -\frac{3}{16}.$$

c) Let

$$F(n) = \frac{n}{n^4+4} \quad \& \quad G(n) = \frac{n}{n^2+9} \quad \text{then}$$

$$F_S(s) = F_S\left(\frac{n}{n^4+4}\right) = \frac{\pi}{2} e^{-2s}.$$

$$G_S(s) = G_S\left(\frac{n}{n^2+9}\right) = \frac{\pi}{2} e^{-3s}.$$

by using Parseval's identity, For Fourier sine transforms.

$$\frac{2}{\pi} \int_0^{\infty} f_c(s) g_c(s) ds = \int_0^{\infty} f(x) g(x) dx$$

$$\Rightarrow \frac{2}{\pi} \int_0^{\infty} \left(\frac{\pi}{2} e^{-2s}\right) \left(\frac{\pi}{2} e^{-3s}\right) ds = \int_0^{\infty} \left(\frac{x}{x^2+4}\right) \left(\frac{x}{x^2+9}\right) dx$$

$$\Rightarrow \frac{\pi}{2} \int_0^{\infty} e^{-5s} ds = \int_0^{\infty} \frac{x^2}{(x^2+4)(x^2+9)} dx$$

$$\Rightarrow \frac{\pi}{2} \left[\frac{e^{-5s}}{-5} \right]_0^{\infty} = \int_0^{\infty} \frac{x^2}{(x^2+4)(x^2+9)} dx$$

$$\Rightarrow \frac{\pi}{2} \left[0 - \frac{1}{5} \right] = \int_0^{\infty} \frac{x^2}{(x^2+4)(x^2+9)} dx$$

$$\frac{\pi}{10} = \int_0^{\infty} \frac{x^2}{(x^2+4)(x^2+9)} dx$$

$$\frac{\pi}{10} = \int_0^{\infty} \frac{t^2}{(t^2+4)(t^2+9)} dt$$

$$\Rightarrow \int_0^{\infty} \frac{t^2}{(t^2+4)(t^2+9)} dt = \frac{\pi}{10}$$

=

4. A] let $u = x + y + z$ & $v = x^2 + y^2 + z^2$, then.

$$F(u, v) = 0$$

diffⁿ partially w.r. to x & y

$$\frac{\partial F}{\partial u} \left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial z} p \right) + \frac{\partial F}{\partial v} \left(\frac{\partial v}{\partial x} + \frac{\partial v}{\partial z} p \right) = 0 \quad \text{--- (1)}$$

$$\Rightarrow \frac{\partial F}{\partial u} (1+p) + \frac{\partial F}{\partial v} (2x+2zp) = 0 \quad \text{--- (1)}$$

$$\& \frac{\partial F}{\partial u} \left(\frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} q \right) + \frac{\partial F}{\partial v} \left(\frac{\partial v}{\partial y} + \frac{\partial v}{\partial z} q \right) = 0$$

$$\Rightarrow \frac{\partial F}{\partial u} (1+q) + \frac{\partial F}{\partial v} (2y+2zq) = 0 \quad \text{--- (2)}$$

Eliminating $\frac{\partial F}{\partial u}$ & $\frac{\partial F}{\partial v}$ from (1) & (2),

$$(1+p)(2y+2zq) = (1+q)(2x+2zp)$$

$$\Rightarrow (y-z)p + (z-x)q = x-y$$

13] The partial differential equation

$$(x^2 - yz)p + (y^2 - zx)q = z^2 - xy \quad \text{--- (1)}$$

is Lagrange's linear of the form.

$$Pp + Qq = R$$

where $P = x^2 - yz$, $Q = y^2 - zx$ & $R = z^2 - xy$.

The Lagrange's auxiliary equations are.

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\text{i.e. } \frac{dx}{x^2-yz} = \frac{dy}{y^2-zx} = \frac{dz}{z^2-xy} \quad \text{--- (2)}$$

From (2) We have

$$\begin{aligned} \frac{dx-dy}{(x^2-yz)-(y^2-zx)} &= \frac{dy-dz}{(y^2-zx)-(z^2-xy)} = \frac{x dx + y dy + z dz}{x^3+y^3+z^3-3xyz} \\ &= \frac{dx+dy+dz}{x^2+y^2+z^2-xy-yz-zx} \quad \text{--- (3)} \end{aligned}$$

From first two ratios of (2)

$$\frac{d(x-y)}{(x-y)(x+y+z)} = \frac{d(y-z)}{(y-z)(x+y+z)}$$

$$\Rightarrow \frac{d(x-y)}{x-y} - \frac{d(y-z)}{y-z} = 0$$

$$\Rightarrow \log(x-y) - \log(y-z) = \log a$$

$$\Rightarrow \frac{x-y}{y-z} = a \quad \text{--- (4)}$$

From the last two ratios of (3)

$$\frac{x dx + y dy + z dz}{(x+y+z)(x^2+y^2+z^2-xy-yz-zx)} = \frac{dx+dy+dz}{x^2+y^2+z^2-xy-yz-zx}$$

$$\Rightarrow \frac{x dx + y dy + z dz}{x+y+z} = d(x+y+z)$$

on int:
$$x dx + y dy + z dz - (x+y+z) d(x+y+z) = 0$$

$$\Rightarrow \frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} - \frac{(x+y+z)^2}{2} = C$$

$$\Rightarrow xy + yz + zx = -C = b \quad \text{--- (5)}$$

From (4) & (5) the general solⁿ is

$$\phi\left(\frac{x-y}{y-z}, xy+yz+zx\right) = 0$$

c) one dimensional heat eqⁿ is

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \quad \text{--- (1)}$$

the solⁿ of (1) consistent with the physical nature of the problem is given by

$$u = C_1 e^{-\frac{h^2 c^2 t}{2}} (C_2 \cosh hx + C_3 \sinh hx) \quad \text{--- (2)}$$

When $u(x, t) = 0$ at $x = 0$ --- (3)

$u(x, t) = 0$ at $x = 2$ --- (4)

$u(x, t) = \sinh \frac{\pi x}{2} + 3 \sinh \frac{5\pi x}{2}$ at $t = 0$ --- (5)

Using the condition (3) in (2), we obtain

$$0 = C_1 e^{-\frac{h^2 c^2 t}{2}} (C_2)$$

$$\Rightarrow C_2 = 0$$

from (2) we have.

$$u = C_1 C_3 e^{-m^2 c^2 t} \sin mx \quad \text{--- (6)}$$

using condition (4) in (2)

$$0 = C_1 C_3 e^{-m^2 c^2 t} \sinh 2m$$

$$\Rightarrow \sinh 2m = 0 = \sinh h\pi$$

$$\Rightarrow 2m = h\pi \quad \Rightarrow m = \frac{h\pi}{2}$$

Now, $u = C_1 C_3 e^{-\left(\frac{h\pi}{2}\right)^2 c^2 t} \sinh \frac{h\pi x}{2} \quad \text{--- (7)}$

The general solⁿ is.

$$u = \sum_{h=1}^{\infty} b_h e^{-\left(\frac{h\pi}{2}\right)^2 c^2 t} \sinh \frac{h\pi x}{2} \quad \text{--- (8)}$$

Using (5) & (8) $\frac{\sinh \pi x}{2} + 3 \frac{\sinh 5\pi x}{2} = \sum_{h=1}^{\infty} b_h \sinh \frac{h\pi x}{2}$

$$= b_1 \frac{\sinh \pi x}{2} + b_2 \frac{\sinh 2\pi x}{2} + b_3 \frac{\sinh 3\pi x}{2} + b_4 \frac{\sinh 4\pi x}{2} + \dots$$

$$\Rightarrow b_1 = 1, b_5 = 3, b_2 = b_3 = b_4 = b_6 = b_7 = 0 = b_8, \dots$$

From (8), we have,

$$u = b_1 e^{-\frac{\pi^2}{4} c^2 t} \sinh \frac{\pi x}{2} + b_5 e^{-\left(\frac{5\pi}{2}\right)^2 c^2 t} \sinh \frac{5\pi x}{2}$$

$$\text{i.e. } u = e^{-\frac{\pi^2}{4} c^2 t} \sinh \frac{\pi x}{2} + 3 e^{-\left(\frac{5\pi}{2}\right)^2 c^2 t} \sinh \frac{5\pi x}{2}$$

Q.E.D.

2.5

Q)

$$u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$$

diffⁿ w.r. to x & y

$$\frac{\partial u}{\partial x} = 3x^2 - 3y^2 + 6x$$

$$\frac{\partial^2 u}{\partial x^2} = 6x + 6$$

$$\frac{\partial u}{\partial y} = -6xy - 6y$$

$$\frac{\partial^2 u}{\partial y^2} = -6x - 6$$

$$\therefore \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$\therefore$ the function u is harmonic

$$\text{Now, } \frac{\partial u}{\partial x} = 3x^2 - 3y^2 + 6x = \frac{\partial v}{\partial y} \text{ (by using C-R eqn)}$$

$$\therefore v = 3xy^2 - y^3 + 6xy + \phi(x)$$

$$\Rightarrow \frac{\partial v}{\partial x} = 6xy + 6y + \phi'(x) = 6xy + 6y \quad \left(\because \frac{\partial u}{\partial y} = \frac{\partial v}{\partial x} \right)$$

$$\Rightarrow \phi'(x) = 0$$

$$\Rightarrow \phi(x) = c$$

$$\therefore v = 3xy^2 - y^3 + 6xy + c$$

$$\text{Hence, } f(z) = u + iv = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1 + i(3xy^2 - y^3 + 6xy + c)$$

B. i) Here the funcⁿ $F(z) = \bar{e}^z$ is an analytic funcⁿ

Also singular point $a = -1$ lies inside the circle
 $|z| = 2$.

by using Cauchy's integral formula,

$$\oint_C \frac{\bar{e}^z}{z+1} dz = 2\pi i F(a) = 2\pi i \bar{e}^{(-1)} \\ = 2\pi i e$$

ii) The funcⁿ $F(z) = \sinh^2 z$ analytic inside & on the circle
 $|z| = 1$ & the singular point $a = \pi/6$ lies inside
the circle.

by using Cauchy's integral formula

$$F''(a) = \frac{2!}{2\pi i} \oint_C \frac{f(z)}{(z-a)^3} dz$$

$$\oint_C \frac{\sinh^2 z}{(z-\pi/6)^3} dz = \pi i \left[\frac{d^2}{dz^2} (\sinh^2 z) \right]_{z=\pi/6}$$

$$= \pi i [2 \cos 2z]_{z=\pi/6}$$

$$= \pi i [2 \cos \pi/3]$$

$$= \pi i (1)$$

$$= \pi i$$

] Here the function $f(z) = \frac{2z-1}{z(z+1)(z-3)}$ has three poles.

$z=0, -1, 3$ of which only $z=0, -1$ lies inside the circle $|z|=2$.

$$\therefore \text{Res}_{z=0} f(z) = \lim_{z \rightarrow 0} z f(z) = \lim_{z \rightarrow 0} \frac{2z-1}{(z+1)(z-3)} = \frac{1}{3}$$

$$\text{Res}_{z=-1} f(z) = \lim_{z \rightarrow -1} (z+1) f(z) = \lim_{z \rightarrow -1} \frac{2z-1}{z(z-3)} = -\frac{3}{4}$$

Hence, by the residue theorem,

$$\begin{aligned} \oint_C \frac{2z-1}{z(z+1)(z-3)} dz &= 2\pi i \{ \text{sum of Residue} \} \\ &= 2\pi i \left(\frac{1}{3} - \frac{3}{4} \right) \\ &= -\frac{5\pi i}{6} \end{aligned}$$